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## Cross-Border Bank Flows and Monetary Policy

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# 1 Big picture

- **Question.** How does monetary policy in source countries affect cross-border bank lending? More specifically, when domestic monetary policy becomes relatively tighter, do banks reduce total lending, or do they reallocate their portfolios across domestic and foreign borrowers?
- **Setting.** The paper studies quarterly cross-border bank claims for many source-destination country pairs from **1995:Q1 to 2014:Q2**. The main data come from the Bank for International Settlements locational banking statistics, combined with newly constructed data on domestic bank claims on nonbank borrowers.
- **Core empirical challenge.** Cross-border flows can change because source-country banks alter credit supply, because destination-country borrowers change credit demand, or because global factors such as the VIX, the dollar, or U.S. monetary policy move all flows at once.
- **Main contribution.** The paper provides multicountry evidence for an international *risk-taking channel* of monetary policy. It does not simply ask whether monetary tightening shrinks lending; it asks how tighter policy changes the *risk composition* of global banks' credit portfolios.
- **Main hypothesis.** A tighter domestic monetary stance makes domestic borrowers relatively riskier by lowering collateral values, income, and net worth. Banks therefore rebalance toward safer borrowers abroad, so cross-border lending grows more than domestic lending.
- **Main empirical design.** The authors exploit the dyadic structure of source-destination bank-flow data. They compare flows from different source countries to the same destination country in the same quarter, using counterparty  $\times$  year-quarter fixed effects to absorb destination-side credit demand.
- **Main baseline result.** A one-percentage-point higher monetary policy rate in the source country raises cross-border credit growth to nonbanks by about **0.53 percentage points**, or roughly **11%** relative to the average cross-border flow to nonbanks. It raises the growth differential between cross-border and domestic nonbank credit by about **0.44 percentage points**, or roughly **15%** of the average differential.
- **Key capital result.** Reallocation toward cross-border lending is stronger for source countries with lower-capitalized or more stressed banking sectors. In the paper's interpretation, this supports a *franchise value* effect more than a pure risk-shifting effect.
- **Key destination-risk result.** Tighter source-country monetary policy reallocates cross-border credit toward safer destinations: advanced economies and countries with investment-grade sovereign ratings. The effect is especially clear for cross-border lending to foreign banks.
- **Main credibility result.** The results survive controls for source-country GDP growth, inflation, credit growth, bank equity returns, sovereign debt, bilateral exchange rates, financial-center status, euro-area status, QE episodes, LIBOR-OIS funding costs, euro-area identification checks, pre/post-GFC splits, and global factors such as the VIX.
- **Economic takeaway.** Monetary policy in one country affects not only domestic credit, but also the international allocation of bank credit. When policy tightens, banks may export credit

toward safer foreign borrowers, so domestic monetary policy can reshape global banking portfolios.

## 2 Data and measurement (institutional context and key objects)

### 2.1 Why cross-border bank flows are the right object

- Global banks lend across borders and therefore can respond to domestic monetary policy by changing the geographic composition of their loan portfolios.
- The relevant question is not only the aggregate amount of cross-border lending, but which source-country banks supply the lending and which destination-country borrowers receive it.
- This makes a bilateral, or dyadic, data set essential: the paper observes flows from source country  $i$  to destination country  $j$  in quarter  $t$ .

**Interpretation.** Aggregate cross-border flows can be dominated by global factors. Dyadic data let the paper separate the source-country supply side from destination-country demand.

### 2.2 Main data source: BIS locational banking statistics

- The main data source is the confidential **BIS locational banking statistics by residence**.
- These data report quarterly aggregate cross-border claims and liabilities of banks residing in reporting countries on counterparties in foreign countries.
- Claims are available by reporting/source country, counterparty/destination country, currency, instrument, and counterparty sector.
- The paper uses BIS-adjusted claims, which are corrected for exchange-rate movements and breaks in series.

**Interpretation.** The key advantage is that the paper can observe who lends to whom, rather than only total capital inflows or outflows.

### 2.3 Sample construction

- The sample period is **1995:Q1–2014:Q2**.
- The final sample contains **29 reporting/source countries** and **76 counterparty/destination countries**.
- Offshore centers are excluded as source countries, except for Hong Kong and Singapore, because offshore-center monetary policy is less relevant for the final allocation of credit.

- Reporting-counterparty pairs with very small outstanding claims, below **\$5 million** in a quarter, are dropped.
- Cross-border claim growth is winsorized at the **2.5th percentile**.

**Interpretation.** The sample balances breadth with comparability. It is large enough to compare many source-country monetary-policy regimes, but screens out observations where tiny claims could generate noisy growth rates.

## 2.4 Domestic credit data

- The BIS locational banking statistics do not provide a historical series of domestic bank claims on domestic borrowers.
- To compare cross-border and domestic credit, the authors build a new data set of bank claims on the domestic nonbank sector.
- Domestic claims combine bank credit to the private nonfinancial sector from the BIS with bank claims, loans, and securities holdings vis-a-vis the public sector from national sources.
- The domestic series includes loans and debt securities, matching the composition of cross-border claims.

**Interpretation.** This construction is central. Without domestic claims, the paper could study cross-border flows, but not portfolio rebalancing between domestic and foreign borrowers.

## 2.5 Dependent variables

- **Cross-border credit flows** are measured as the growth rate of cross-border bank claims:

$$\text{Credit flow}_{ijt} = \frac{\Delta \text{Claims}_{ijt}}{\text{Claims}_{ij,t-1}}.$$

- The paper separately studies cross-border flows to all sectors, to banks, and to nonbanks.
- **Domestic credit flows** are the growth rate of domestic bank claims on domestic nonbank borrowers.
- **Credit growth differential** is cross-border nonbank credit growth minus domestic nonbank credit growth.

**Interpretation.** The dependent variables are not dollar flows. They are normalized by lagged claims, so the coefficients are percentage-point changes in credit growth.

## 2.6 Main explanatory variable: source-country policy rate

- The main explanatory variable is the lagged nominal monetary policy rate in the source country.

- The paper uses the *level* of the policy rate rather than the change in the policy rate.
- For euro-area countries, the paper uses each country's own policy rate before the euro and the ECB main refinancing operations rate after the introduction of the euro.

**Interpretation.** The level of the policy rate captures the relative stance of monetary policy across source countries in the same quarter. The use of lags also makes the timing more consistent with a policy-to-lending channel.

## 2.7 Source-country controls

- Real GDP growth.
- Inflation.
- Domestic credit growth.
- Sovereign debt-to-GDP ratio.
- Bank equity returns.
- Bilateral exchange-rate changes.
- Financial-center indicator.
- Euro-area indicator.
- QE indicator for Japan, the United Kingdom, and the United States after policy rates hit the effective lower bound.

**Interpretation.** These controls are designed to reduce the concern that monetary policy merely proxies for domestic macroeconomic or financial conditions in source countries.

## 2.8 Banking-sector capital and risk measures

- **High SRISK:** equals one when the source country's banking-sector SRISK-to-GDP ratio is above the sample median in a given year.
- **Low capital:** equals one when the banking sector's equity-capital-to-assets ratio is below the sample median in a given year.
- **High housing:** equals one when residential mortgage loans are a high share of total domestic credit.

**Interpretation.** These variables test whether bank balance-sheet strength changes the response to monetary policy. High SRISK, low capital, and high mortgage exposure are all related to lower resilience or greater interest-rate sensitivity.

## 2.9 Destination-country risk measures

- **Speculative-grade counterparty:** equals one if the destination country has a non-investment-grade sovereign rating in a given quarter.
- **Emerging-market counterparty:** equals one if the destination country is classified as an emerging market economy.

**Interpretation.** These variables convert the portfolio-channel prediction into a testable cross-sectional implication: when source-country policy tightens, flows should shift toward safer destinations.

## 2.10 Descriptive facts

- Average quarterly cross-border credit growth to all sectors is about **4.0%**.
- Average quarterly cross-border credit growth to banks is about **8.9%**, much more volatile than flows to nonbanks.
- Average quarterly cross-border credit growth to nonbanks is about **4.7%**.
- Average quarterly domestic credit growth to nonbanks is only about **1.8%**.
- The mean growth differential between cross-border and domestic nonbank credit is about **3.1 percentage points**.

**Interpretation.** Cross-border credit is both faster-growing and more volatile than domestic credit, which makes portfolio reallocation quantitatively important.

# 3 Models and empirical specifications (all equations + interpretation)

## 3.1 Conceptual framework: international risk-taking channel

- The paper distinguishes the **bank lending channel** from the **risk-taking channel**.
- The bank lending channel concerns the total quantity of credit: tighter policy raises funding costs and may reduce aggregate lending.
- The risk-taking channel concerns the composition of credit: tighter policy changes perceived borrower risk and banks' risk tolerance.
- The paper focuses on the risk-taking channel in an international setting.

**Interpretation.** The paper's key prediction is compositional. Tighter source-country monetary policy can reduce domestic lending while increasing cross-border lending to safer foreign borrowers.

### 3.2 Baseline stacked domestic/cross-border regression

The first specification stacks cross-border and domestic nonbank credit flows:

$$\text{CreditFlow}_{ijt} = \alpha \text{PolicyRate}_{i,t-1}^{rep} + \gamma \text{Domestic}_i + \beta \text{PolicyRate}_{i,t-1}^{rep} \times \text{Domestic}_i + \delta' X_{i,t-1}^{rep} + \zeta_{jt} + \epsilon_{ijt}. \quad (1)$$

Here:

- $\text{CreditFlow}_{ijt}$  is credit growth from source country  $i$  to destination country  $j$  in quarter  $t$ ;
- $\text{Domestic}_i = 1$  for domestic nonbank credit and 0 for cross-border nonbank credit;
- $X_{i,t-1}^{rep}$  are source-country controls;
- $\zeta_{jt}$  are counterparty  $\times$  year-quarter fixed effects.

**Interpretation.** The coefficient  $\alpha$  gives the effect of source-country policy on cross-border credit growth. The sum  $\alpha + \beta$  gives the effect on domestic credit growth. A negative  $\beta$  means domestic lending responds less positively than cross-border lending, consistent with international portfolio rebalancing.

### 3.3 Credit-growth differential regression

The second specification directly measures the gap between cross-border and domestic nonbank credit growth:

$$\text{CreditGrowthDiff}_{ijt} = \alpha \text{PolicyRate}_{i,t-1}^{rep} + \delta' X_{i,t-1}^{rep} + \zeta_{jt} + \epsilon_{ijt}. \quad (2)$$

**Interpretation.** A positive  $\alpha$  means that when the source-country policy rate is higher, cross-border nonbank credit grows faster than domestic nonbank credit. This is the cleanest reduced-form test of portfolio reallocation.

### 3.4 Bank capital interaction

The third specification asks whether reallocation is stronger for source countries with weaker or more exposed banking sectors:

$$\text{CreditGrowthDiff}_{ijt} = \alpha \text{PolicyRate}_{i,t-1}^{rep} + \beta \text{PolicyRate}_{i,t-1}^{rep} \times \text{Banking}_{i,t-1}^{rep} + \gamma \text{Banking}_{i,t-1}^{rep} + \delta' X_{i,t-1}^{rep} + \zeta_{jt} + \epsilon_{ijt}. \quad (3)$$

**Interpretation.**  $\text{Banking}_{i,t-1}^{rep}$  is alternatively High SRISK, Low capital, or High housing. A positive  $\beta$  means that banks in weaker or more rate-sensitive banking systems rebalance more toward foreign borrowers when monetary policy tightens.

### 3.5 Risk-shifting versus franchise value

- **Risk-shifting prediction.** Lower-capital banks may tolerate more risk because limited liability protects shareholders from full downside losses. Under this view, low-capital banks should



rebalance less toward safer foreign borrowers.

- **Franchise value prediction.** Lower-capital banks may have higher franchise values and stronger incentives to preserve future rents. Under this view, low-capital banks should rebalance more toward safer foreign borrowers.
- The sign of  $\beta$  in the capital-interaction regression distinguishes these forces empirically.

**Interpretation.** The paper finds positive interactions for High SRISK, Low capital, and High housing, so it interprets the evidence as more consistent with franchise value incentives than with pure risk shifting.

### 3.6 Destination-risk interaction

The fourth specification tests whether cross-border flows go to safer destinations:

$$\begin{aligned} \text{CrossBorderFlow}_{ijt} = & \alpha \text{PolicyRate}_{i,t-1}^{rep} + \beta \text{PolicyRate}_{i,t-1}^{rep} \times \text{Risk}_{j,t-1}^{cp} \\ & + \gamma' X_{i,t-1}^{rep} + \delta' X_{i,t-1}^{rep} \times \text{Risk}_{j,t-1}^{cp} + \eta \text{Risk}_{j,t-1}^{cp} + \zeta_{jt} + \epsilon_{ijt}. \end{aligned} \quad (4)$$

Here  $\text{Risk}^{cp}$  is either a speculative-grade sovereign-rating indicator or an emerging-market indicator.

**Interpretation.** A negative  $\beta$  means that the positive effect of tight source-country policy on cross-border lending is weaker for risky destinations. This supports the view that tighter policy pushes banks toward safer foreign borrowers.

### 3.7 Global-factor robustness specification

The paper also estimates a specification that nests source-country monetary policy with destination-country conditions and global factors:

$$\text{CrossBorderFlow}_{ijt} = \alpha \text{PolicyRate}_{i,t-1}^{rep} + \theta \text{PolicyRate}_{j,t-1}^{cp} + \beta' X_{i,t-1}^{rep} + \mu' Y_{j,t-1}^{cp} + \gamma_{ij} + \phi_t + \epsilon_{ijt}. \quad (5)$$

**Interpretation.** This specification uses reporting-counterparty fixed effects and time fixed effects, or replaces time fixed effects with explicit global factors such as  $\ln(\text{VIX})$ . The key question is whether source-country policy still matters after global financial conditions are absorbed.

### 3.8 Fixed effects and standard errors

- The main specifications include counterparty  $\times$  year-quarter fixed effects.
- These fixed effects absorb time-varying demand conditions in each destination country.
- Standard errors are double-clustered by reporting/source country and counterparty/destination country.

**Interpretation.** The paper's identification uses variation across source countries lending to the same destination country in the same quarter. The clustering accounts for serial correlation and cross-flow dependence at both ends of the dyad.

## 4 Identification and empirical design (what makes the estimates credible)

### 4.1 What identifies the main estimate

- The main source of identifying variation is the difference in monetary policy stances across source countries within the same destination country and quarter.
- Counterparty  $\times$  year-quarter fixed effects compare lending from different source countries to the same destination country at the same time.
- This means destination-specific credit demand, local macro shocks, and recipient-country financial conditions are absorbed.

**Economic message.** The estimate is designed to isolate supply-side changes in cross-border lending that originate from source-country monetary policy.

### 4.2 Why the dyadic design matters

- Balance-of-payments flows usually reveal only aggregate inflows or outflows.
- BIS dyadic data reveal a matrix of flows: source country  $i$  to destination country  $j$  in quarter  $t$ .
- Because many source countries lend to the same destination in the same quarter, the paper can control for destination-time demand shocks.

**Economic message.** The dyadic design is the paper's core empirical advantage. It makes the paper less vulnerable to the standard problem that capital flows confound supply and demand.

### 4.3 Addressing source-country endogeneity

- Monetary policy is not randomly assigned. It responds to domestic macroeconomic and financial conditions.
- The paper controls for source-country GDP growth, inflation, credit growth, sovereign debt, bank equity returns, exchange-rate movements, financial-center status, euro-area status, and QE periods.
- It also performs a euro-area robustness test, where a common ECB policy rate may be less tightly tied to each individual member country's conditions.

**Economic message.** The paper cannot claim a pure randomized policy shock, but it builds a strong case that the estimated source-policy effect is not merely a proxy for domestic booms or recessions.

## 4.4 Domestic versus cross-border bank overlap

- The data are aggregated at the country level, so the exact set of banks making domestic loans may not perfectly match the set making cross-border loans.
- The authors address this concern by considering banking-system concentration and related robustness checks.
- The main results remain similar, and the effects are slightly stronger for more concentrated banking sectors, where large banks are more likely to operate both domestically and internationally.

**Economic message.** Imperfect bank overlap may add noise, but the evidence does not suggest that it drives the result.

## 4.5 Separating risk-taking from bank lending

- The bank lending channel predicts changes in the total quantity of credit.
- The risk-taking channel predicts changes in the composition of credit across safer and riskier borrowers.
- The paper’s strongest evidence comes from compositional patterns: cross-border versus domestic, safer versus riskier destination countries, and lower-capital versus higher-capital banking systems.

**Economic message.** The results are not just “tighter policy changes lending.” The evidence points to portfolio reallocation toward safer foreign exposure.

## 4.6 Global factors

- Global bank flows are influenced by U.S. monetary policy, the VIX, the dollar, and global liquidity.
- The paper controls for these forces using year-quarter fixed effects, explicit VIX controls, broad dollar robustness checks, and policy-rate differentials.
- The source-country policy coefficient remains positive in the global-factor specifications.

**Economic message.** The paper argues that source-country monetary policy is not simply a disguised global financial cycle. Local policy stances matter at the same time as global factors.

## 4.7 What the paper can and cannot show

- The paper can show country-level portfolio reallocation patterns across domestic and cross-border credit.
- It can show that those patterns are consistent with the risk-taking channel.

- It cannot observe loan-level borrower risk, individual bank decisions, or individual bank balance sheets for every country-pair exposure.
- It also cannot fully randomize monetary policy across countries.

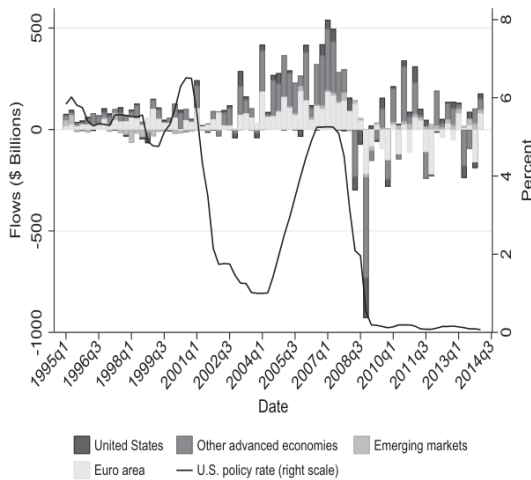
**Economic message.** The paper’s evidence is strongest as a macro-finance, country-pair-level test of an international portfolio channel.

## 5 Tables and figures

### 5.1 Figure 1: composition of cross-border flows and the U.S. policy rate

**Read:** Figure 1 plots four-quarter cumulative cross-border bank flows by source-region group and overlays the U.S. policy rate. The bars show that the sources of global bank credit change substantially over time, especially around the global financial crisis.

**Economic message.** Prior work often studies the height of the total bar, meaning aggregate global flows. This paper studies the composition of the bar: which source-country banking systems are expanding or contracting cross-border credit in response to their own monetary policies.



**Figure 1**  
**Composition of cross-border bank flows and the U.S. policy rate**  
The figure reports quarterly data between 1995:Q1 and 2014:Q1 for countries that report cross-border bank claims to the Bank for International Settlements for the entire sample period. The bars break- and exchange rate-adjusted cross-border flows for reporting countries aggregated into four regions: the United States, the euro area, other advanced economies, and emerging markets. The *euro area* comprises Austria, Belgium, Finland, France, Germany, Ireland, Italy, Luxembourg, Netherlands, and Spain. *Other advanced economies* include Canada, Denmark, Japan, Norway, Sweden, Switzerland, and the United Kingdom. Lastly, the *emerging markets* group is made up of Hong Kong and Singapore. The solid line represents the effective federal funds rate.

**Table 1**  
**Summary statistics**

|                                                     | Mean   | Std. dev. | 25th    | 50th   | 75th   |
|-----------------------------------------------------|--------|-----------|---------|--------|--------|
| <b>All sample countries</b>                         |        |           |         |        |        |
| Credit flows, domestic and cross-border (%)         | 4.539  | 26.306    | -6.605  | 0.653  | 9.998  |
| Cross-border credit flows: All sectors (%)          | 3.971  | 24.034    | -7.587  | 0.837  | 11.089 |
| Cross-border credit flows: Banks (%)                | 8.877  | 46.754    | -12.616 | 0.495  | 16.955 |
| Cross-border credit flows: Nonbanks (%)             | 4.675  | 26.757    | -6.872  | 0.545  | 10.378 |
| Domestic credit flows to nonbanks (%)               | 1.800  | 3.395     | 0.324   | 1.504  | 2.841  |
| Credit growth diff., cross-border vs. domestic (pp) | 3.068  | 26.089    | -8.047  | -0.222 | 8.673  |
| <b>Reporting countries (rep)</b>                    |        |           |         |        |        |
| Bank equity returns rep (% yoy)                     | 2.523  | 17.283    | -5.335  | 3.016  | 11.082 |
| Capital/assets rep (%)                              | 6.095  | 2.441     | 4.386   | 5.767  | 7.434  |
| Credit growth rep (% yoy)                           | 7.814  | 12.802    | -0.66   | 6.635  | 15.186 |
| Debt/GDP rep (%)                                    | 69.246 | 39.931    | 46.52   | 63.32  | 82.65  |
| EME rep                                             | 0.063  | 0.243     | 0       | 0      | 0      |
| Euro area rep                                       | 0.447  | 0.497     | 0       | 0      | 1      |
| High Housing rep                                    | 0.613  | 0.487     | 0       | 1      | 1      |
| High SRISK rep                                      | 0.592  | 0.491     | 0       | 1      | 1      |
| Housing/GDP rep                                     | 0.448  | 0.246     | 0.244   | 0.413  | 0.627  |
| Financial center rep                                | 0.19   | 0.392     | 0       | 0      | 0      |
| Inflation rep (% yoy)                               | 0.523  | 0.613     | 0.198   | 0.481  | 0.801  |
| Low capital rep                                     | 0.388  | 0.487     | 0       | 1      | 1      |
| QE indicator rep                                    | 0.057  | 0.233     | 0       | 0      | 0      |
| Policy rate rep (%)                                 | 2.796  | 2.46      | 1       | 2.5    | 4      |
| Shadow policy rate rep (%)                          | 2.321  | 3.023     | 0.446   | 2.451  | 3.920  |
| Real GDP growth rep (% yoy)                         | 0.522  | 1.023     | 0.089   | 0.569  | 1.012  |
| SRISK/GDP rep                                       | 0.051  | 0.052     | 0.008   | 0.037  | 0.08   |
| <b>Counterparty countries (cp)</b>                  |        |           |         |        |        |
| Bank equity returns cp (% yoy)                      | 3.522  | 19.204    | -5.426  | 3.173  | 12.232 |
| Credit growth cp (% yoy)                            | 9.89   | 14.838    | 0.594   | 8.389  | 18.303 |
| Debt/GDP cp (%)                                     | 56.244 | 35.659    | 33.88   | 48.18  | 72.32  |
| EME cp                                              | 0.558  | 0.497     | 0       | 1      | 1      |
| Inflation cp (% yoy)                                | 1.099  | 4.747     | 0.244   | 0.659  | 1.248  |
| Policy rate cp (%)                                  | 5.207  | 7.439     | 1.83    | 3.69   | 6.03   |
| Real GDP growth cp (% yoy)                          | 0.72   | 1.388     | 0.118   | 0.752  | 1.391  |
| Speculative-grade cp                                | 0.208  | 0.406     | 0       | 0      | 0      |
| <b>Other variables</b>                              |        |           |         |        |        |
| Exchange rate pct. change (% yoy)                   | 1.154  | 9.237     | -3.351  | 0      | 5.228  |
| LIBOR-OIS spread rep                                | 0.169  | 0.343     | 0.600   | 0.080  | 0.250  |
| ln(VIX)                                             | 3.027  | 0.345     | 2.761   | 3.065  | 3.261  |
| Policy rate differential (pp)                       | -2.443 | 7.657     | -3.75   | -0.75  | 0.25   |
| U.S. policy rate (%)                                | 2.799  | 2.279     | 0.19    | 2.12   | 5.25   |

This table reports summary statistics for the main dependent variables, as well as for the explanatory variables in reporting (source) and counterparty (destination) countries. The sample period is from 1995:Q1 to 2014:Q2. The sample includes all reporting and counterparty countries listed in Table A1 in the appendix. The appendix defines all variables.

## 5.2 Table 1: summary statistics

**Read:** Table 1 reports the main dependent and explanatory variables. Cross-border credit growth to nonbanks averages 4.675%, while domestic credit growth to nonbanks averages 1.800%. Source-country policy rates average 2.796%, and counterparty-country policy rates average 5.207%.

**Economic message.** Cross-border credit is larger and more volatile than domestic credit. This creates room for monetary policy to affect not only credit quantities but also the international composition of banks' portfolios.

## 5.3 Table 2: portfolio reallocation between domestic and cross-border credit

**Read:** The full-sample coefficient on the lagged source-country policy rate is 0.527 for cross-border/domestic stacked credit flows and 0.440 for the cross-border-minus-domestic growth differential. The interaction with the domestic-credit indicator is  $-0.398$ , implying that domestic credit responds less than cross-border credit.

**Economic message.** When monetary policy is tighter in the source country, banks tilt away from domestic nonbank credit and toward cross-border nonbank credit. This is the paper's central portfolio-reallocation result.

Table 2  
Portfolio reallocation between domestic and cross-border credit

|                                       | Full sample                             |                                                | Pre-GFC (until 2007:Q2)                 |                                                | Post-GFC (2009:Q3 onward)               |                                                |
|---------------------------------------|-----------------------------------------|------------------------------------------------|-----------------------------------------|------------------------------------------------|-----------------------------------------|------------------------------------------------|
|                                       | Credit flows, domestic and cross-border | Credit growth diff., cross-border vs. domestic | Credit flows, domestic and cross-border | Credit growth diff., cross-border vs. domestic | Credit flows, domestic and cross-border | Credit growth diff., cross-border vs. domestic |
|                                       | (1)                                     | (2)                                            | (3)                                     | (4)                                            | (5)                                     | (6)                                            |
| Lag policy rate rep                   | 0.527***<br>[0.120]                     | 0.440**<br>[0.113]                             | 0.548**<br>[0.190]                      | 0.400**<br>[0.189]                             | 0.682**<br>[0.315]                      | 0.637*<br>[0.311]                              |
| Lag policy rate $\times$ Domestic     | -0.398***<br>[0.097]                    |                                                | -0.471**<br>[0.181]                     |                                                | -0.372<br>[0.254]                       |                                                |
| Domestic                              | -2.266***<br>[0.431]                    |                                                | -3.329***<br>[0.637]                    |                                                | -1.264*<br>[0.679]                      |                                                |
| Lag bank equity returns rep           | -0.011<br>[0.010]                       | -0.020**<br>[0.009]                            | -0.007<br>[0.015]                       | -0.022<br>[0.013]                              | -0.029<br>[0.021]                       | -0.019<br>[0.021]                              |
| Lag real GDP growth rep               | 0.479***<br>[0.184]                     | 0.307<br>[0.220]                               | 0.018<br>[0.230]                        | -0.143<br>[0.267]                              | 1.136***<br>[0.266]                     | 0.922***<br>[0.331]                            |
| Lag debt/GDP rep                      | -0.005<br>[0.005]                       | 0.008<br>[0.006]                               | 0.005<br>[0.006]                        | 0.019***<br>[0.007]                            | -0.002<br>[0.005]                       | 0.003<br>[0.006]                               |
| Lag inflation rep                     | 0.046<br>[0.121]                        | -0.079<br>[0.120]                              | 0.221<br>[0.136]                        | 0.052<br>[0.138]                               | 0.041<br>[0.120]                        | -0.008<br>[0.133]                              |
| Exchange rate pct. change             | 0.029<br>[0.023]                        | 0.043<br>[0.020]                               | 0.011<br>[0.039]                        | 0.027<br>[0.040]                               | -0.007<br>[0.040]                       | 0.040<br>[0.045]                               |
| Financial center rep                  | -1.386<br>[0.906]                       | -1.172<br>[0.879]                              | -1.577*<br>[0.811]                      | -1.016<br>[0.767]                              | 1.409*<br>[0.771]                       | 0.888<br>[0.918]                               |
| Euro area rep                         | -1.666***<br>[0.541]                    | -1.378***<br>[0.464]                           | -1.581**<br>[0.612]                     | -1.442***<br>[0.510]                           | -2.116***<br>[0.561]                    | -1.302**<br>[0.569]                            |
| QE indicator rep                      | 1.215<br>[0.874]                        | 1.060<br>[1.174]                               | 1.241<br>[0.776]                        | 1.019<br>[0.742]                               | -1.645<br>[1.381]                       | -0.169<br>[1.549]                              |
| EMI rep                               | -0.083<br>[0.980]                       | -0.895<br>[0.979]                              | 0.133<br>[1.330]                        | 0.572<br>[1.177]                               | -1.710<br>[1.283]                       | -2.979**<br>[1.437]                            |
| Counterparty $\times$ year-quarter FE | Yes                                     | Yes                                            | Yes                                     | Yes                                            | Yes                                     | Yes                                            |
| Observations                          | 66,796                                  | 65,202                                         | 38,828                                  | 36,085                                         | 21,105                                  | 20,559                                         |
| R <sup>2</sup>                        | .11                                     | .11                                            | .11                                     | .12                                            | .10                                     | .09                                            |

The dependent variable in columns 1, 3, and 5 combines the domestic and cross-border credit flows to nonbank borrowers. The dependent variable in columns 2, 4, and 6 is the credit growth differential between cross-border and domestic nonbank borrowers. *Domestic* is an indicator variable that equals one for domestic credit and zero for cross-border credit. The sample period in columns 1 and 2 is from 1995:Q1 to 2004:Q2, from 1995:Q1 to 2007:Q2 in columns 3 and 4, and from 2009:Q3 to 2014:Q2 in columns 5 and 6. The sample includes all reporting and counterparty countries listed in Table A1 in the appendix. The appendix defines all variables. Standard errors are clustered at the counterparty and reporting country levels. \* $p < .1$ ; \*\* $p < .05$ ; \*\*\* $p < .01$ .

Table 3  
Cross-border portfolio reallocation and bank capital in source countries

|                                           | Credit growth differential, cross-border vs. domestic |                      |                      |
|-------------------------------------------|-------------------------------------------------------|----------------------|----------------------|
|                                           | (1)                                                   | (2)                  | (3)                  |
| Lag policy rate rep                       | 0.414***<br>[0.100]                                   | 0.465***<br>[0.112]  | 0.359**<br>[0.130]   |
| High SRISK rep                            | -2.365***<br>[0.694]                                  |                      |                      |
| Lag policy rate $\times$ High SRISK rep   | 0.556**<br>[0.252]                                    |                      |                      |
| Low capital rep                           |                                                       | -1.118**<br>[0.432]  |                      |
| Lag policy rate $\times$ Low capital rep  |                                                       | 0.415**<br>[0.169]   |                      |
| High Housing rep                          |                                                       |                      | -0.949*<br>[0.549]   |
| Lag policy rate $\times$ High housing rep |                                                       |                      | 0.314*<br>[0.175]    |
| Lag bank equity returns rep               | -0.024**<br>[0.011]                                   | -0.025**<br>[0.009]  | -0.014**<br>[0.006]  |
| Lag real GDP growth rep                   | 0.264<br>[0.241]                                      | 0.311<br>[0.228]     | 0.171<br>[0.163]     |
| Lag debt/GDP rep                          | 0.008<br>[0.005]                                      | 0.010*<br>[0.006]    | 0.005<br>[0.006]     |
| Lag inflation rep                         | -0.138<br>[0.117]                                     | -0.108<br>[0.116]    | -0.106<br>[0.074]    |
| Exchange rate pct. change                 | 0.043<br>[0.034]                                      | 0.064*<br>[0.032]    | 0.013<br>[0.025]     |
| Financial center rep                      | -1.517**<br>[0.579]                                   | -1.313*<br>[0.741]   | -0.029<br>[0.808]    |
| Euro area rep                             | -1.687***<br>[0.419]                                  | -1.606***<br>[0.432] | -1.357***<br>[0.464] |
| QE indicator rep                          | 1.388<br>[0.917]                                      | 1.132<br>[1.292]     | 1.479<br>[0.952]     |
| EME rep                                   | -1.395*<br>[0.819]                                    | -1.154<br>[0.917]    | 0.609<br>[1.269]     |
| Counterparty $\times$ Year-quarter FE     | Yes                                                   | Yes                  | Yes                  |
| Observations                              | 53,644                                                | 61,378               | 51,098               |
| R <sup>2</sup>                            | .10                                                   | .13                  | .13                  |

The dependent variable is the credit growth differential between cross-border and domestic nonbank borrowers. *HighSRISK rep* is an indicator variable that takes the value of one for values higher than the median value in a given year, and zero otherwise. *Low capital rep* is an indicator variable that takes the value of one for values lower than the median Tier 1 capital ratio in a given year, and zero otherwise. *High housing rep* is an indicator variable that takes the value of one for values higher than the median share of housing credit in GDP in a given year, and zero otherwise. The sample period is from 1995:Q1 to 2014:Q2. The sample includes all reporting and counterparty countries listed in Table A1 in the appendix. The appendix defines all variables. Standard errors are clustered at the counterparty and reporting country levels. \* $p < .1$ ; \*\* $p < .05$ ; \*\*\* $p < .01$ .

## 5.4 Table 3: source-country bank capital and portfolio reallocation

**Read:** Table 3 interacts source-country policy rates with measures of banking-sector weakness or exposure. The interaction is positive for High SRISK (0.556), Low capital (0.415), and High housing exposure (0.314).

**Economic message.** Reallocation is stronger for banking systems that are lower-capitalized, more systemically risky, or more exposed to housing. This supports the paper's interpretation that franchise-value incentives dominate simple limited-liability risk shifting in the sample.

## 5.5 Table 4: cross-border credit flows and source-country monetary policy

**Read:** Table 4 studies cross-border flows only. A one-percentage-point higher source-country policy rate is associated with higher cross-border credit growth to all sectors (0.312), banks (0.252), and nonbanks (0.463).

**Economic message.** The positive source-policy effect is not limited to the domestic-versus-foreign comparison. Even within cross-border flows alone, tighter source-country policy is associated with more lending abroad.

**Table 4**  
Cross-border credit flows and monetary policy in source countries

|                                | Cross-border credit flows to |                      |                      |
|--------------------------------|------------------------------|----------------------|----------------------|
|                                | All sectors                  | Banks                | Nonbanks             |
|                                | (1)                          | (2)                  | (3)                  |
| Lag policy rate rep            | 0.312***<br>[0.083]          | 0.252**<br>[0.093]   | 0.463***<br>[0.132]  |
| Lag credit growth rep          | 0.058*<br>[0.029]            | 0.086*<br>[0.050]    | 0.060*<br>[0.031]    |
| Lag bank equity returns rep    | -0.005<br>[0.011]            | -0.000<br>[0.021]    | -0.013<br>[0.010]    |
| Lag real GDP growth rep        | 0.262**<br>[0.111]           | 0.175<br>[0.249]     | 0.399**<br>[0.170]   |
| Lag debt/GDP rep               | -0.006<br>[0.004]            | -0.023***<br>[0.008] | -0.004<br>[0.006]    |
| Lag inflation rep              | -0.028<br>[0.305]            | 0.802<br>[0.603]     | 0.051<br>[0.525]     |
| Exchange rate pct. change      | -0.012<br>[0.026]            | 0.015<br>[0.046]     | -0.013<br>[0.031]    |
| Financial center rep           | -1.301*<br>[0.677]           | -3.602***<br>[0.989] | -1.530*<br>[0.818]   |
| Euro area rep                  | -1.093**<br>[0.397]          | -1.705*<br>[0.980]   | -1.695***<br>[0.538] |
| QE indicator rep               | 1.542***<br>[0.460]          | 1.299<br>[1.076]     | 1.609*<br>[0.915]    |
| EME rep                        | 0.221<br>[0.992]             | 1.479<br>[1.655]     | 0.278<br>[1.154]     |
| Counterparty × Year-quarter FE | Yes                          | Yes                  | Yes                  |
| Observations                   | 68,403                       | 66,170               | 67,018               |
| R <sup>2</sup>                 | .12                          | .12                  | .11                  |

The dependent variable is the cross-border credit flows to borrowers in all sectors, to banks, or to nonbanks. The sample period is from 1995:Q1 to 2014:Q2. The sample includes all reporting and counterparty countries listed in Table A1 in the appendix. The appendix defines all variables. Standard errors are clustered at the counterparty and reporting country levels. \*  $p < .1$ ; \*\*  $p < .05$ ; \*\*\*  $p < .01$ .

**Table 5**  
Cross-border credit flows and counterparty risk in destination countries

|                                            | Cross-border credit flows to |                     |                    |
|--------------------------------------------|------------------------------|---------------------|--------------------|
|                                            | All sectors                  | Banks               | Nonbanks           |
|                                            | (1)                          | (2)                 | (3)                |
| <i>A. Speculative-grade counterparties</i> |                              |                     |                    |
| Lag policy rate rep                        | 0.352***<br>[0.072]          | 0.413***<br>(0.065) | 0.354**<br>[0.160] |
| Lag policy rate rep × Spec.-grade cp       | -0.277**<br>[0.133]          | -0.545**<br>(0.243) | 0.049<br>[0.203]   |
| <i>Joint speculative grade cp</i>          | 0.0751                       | -0.132              | 0.403              |
| <i>t</i> -statistic                        | 0.477                        | -0.487              | 2.782              |
| Controls rep                               | Yes                          | Yes                 | Yes                |
| Controls rep × Spec.-grade cp              | Yes                          | Yes                 | Yes                |
| Counterparty × Year-quarter FE             | Yes                          | Yes                 | Yes                |
| Observations                               | 68,403                       | 66,170              | 67,018             |
| R <sup>2</sup>                             | .11                          | .11                 | .10                |
| <i>B. Emerging market counterparties</i>   |                              |                     |                    |
| Lag policy rate rep                        | 0.375***<br>[0.072]          | 0.492***<br>[0.063] | 0.275*<br>[0.160]  |
| Lag policy rate rep × EME cp               | -0.257*<br>[0.129]           | -0.682**<br>[0.281] | 0.258<br>[0.255]   |
| <i>Joint EME cp</i>                        | 0.118                        | -0.190              | 0.533              |
| <i>t</i> -statistic                        | 0.690                        | -0.731              | 2.840              |
| Controls rep                               | Yes                          | Yes                 | Yes                |
| Controls rep × EME cp                      | Yes                          | Yes                 | Yes                |
| Counterparty × Year-quarter FE             | Yes                          | Yes                 | Yes                |
| Observations                               | 68,403                       | 66,170              | 67,018             |
| R <sup>2</sup>                             | .12                          | .12                 | .11                |

The dependent variable is the cross-border credit flows to borrowers in all sectors, to banks, and to nonbanks. The indicator variables *Speculative grade cp* and *EME cp* equal one if the counterparty country has a non-investment-grade rating and EME status, respectively, in a given year-quarter. The sum of the coefficients for *Lag policy rate rep* and *Lag policy rate rep × Speculative grade cp* is reported as *Joint speculative grade cp* with the corresponding *t*-statistic. The sum of coefficients for *Lag policy rate rep* and *Lag policy rate rep × EME cp* is reported as *Joint EME cp*. The sample period is from 1995:Q1 to 2014:Q2. The sample includes all reporting and counterparty countries listed in Table A1 in the appendix. The appendix defines all variables. Standard errors are

## 5.6 Table 5: counterparty risk in destination countries

**Read:** Table 5 interacts the source-country policy rate with risky-destination indicators. The interaction is negative for speculative-grade counterparties in all-sector and bank flows, and negative for emerging-market counterparties in all-sector and bank flows.

**Economic message.** Banks do not simply send more credit abroad indiscriminately. Under tighter domestic monetary policy, cross-border bank credit shifts toward safer foreign destinations.

## 5.7 Table 6: banks' cost of funding

**Read:** Table 6 adds the source-country LIBOR-OIS spread as a proxy for bank funding costs. The source policy-rate coefficient remains positive for the growth differential and for cross-border flows to all sectors and nonbanks. The LIBOR-OIS coefficient is negative in the growth-differential and nonbank-flow columns.

**Table 6**  
Cross-border credit flows and banks' cost of funding

|                                | Credit growth diff.,<br>cross-border vs.<br>domestic | Cross-border credit flows to |                      |                      |
|--------------------------------|------------------------------------------------------|------------------------------|----------------------|----------------------|
|                                |                                                      | All sectors                  | Banks                | Nonbanks             |
|                                | (1)                                                  | (2)                          | (3)                  | (4)                  |
| Lag policy rate rep            | 0.402**<br>[0.192]                                   | 0.275**<br>[0.125]           | 0.145<br>[0.229]     | 0.340*<br>[0.168]    |
| Lag LIBOR-OIS rep              | -0.968*<br>[0.532]                                   | -0.495<br>[0.416]            | -0.986<br>[0.736]    | -0.724*<br>[0.369]   |
| Lag credit growth rep          |                                                      | 0.046<br>[0.029]             | 0.062<br>[0.052]     | 0.060<br>[0.036]     |
| Lag bank equity returns rep    | -0.027**<br>[0.010]                                  | -0.003<br>[0.010]            | -0.025**<br>[0.023]  |                      |
| Lag real GDP growth rep        | 0.381<br>[0.274]                                     | 0.374***<br>[0.126]          | 0.172<br>[0.234]     | 0.462**<br>[0.194]   |
| Lag debt/GDP rep               | 0.009<br>[0.008]                                     | -0.004<br>[0.005]            | -0.021**<br>[0.009]  | -0.005<br>[0.007]    |
| Lag inflation rep              | 0.005<br>[0.104]                                     | 0.104<br>[0.272]             | 0.975<br>[0.640]     | 0.534<br>[0.455]     |
| Exchange rate                  | 0.037<br>[0.035]                                     | -0.000<br>[0.028]            | 0.038<br>[0.047]     | -0.038<br>[0.040]    |
| Financial center               | -0.977<br>[0.948]                                    | -1.026<br>[0.635]            | -3.154***<br>[1.048] | -1.420*<br>[0.813]   |
| Euro area rep                  | -1.221**<br>[0.572]                                  | -0.832*<br>[0.423]           | -1.365<br>[1.028]    | -1.600***<br>[0.564] |
| QE indicator rep               | 0.973<br>[1.242]                                     | 1.378**<br>[0.494]           | 1.077<br>[1.019]     | 1.606<br>[1.009]     |
| EME rep                        | -0.618<br>[2.196]                                    | 0.296<br>[0.962]             | 2.670<br>[2.543]     | 0.966<br>[1.555]     |
| Counterparty × Year-quarter FE | Yes                                                  | Yes                          | Yes                  | Yes                  |
| Observations                   | 58,123                                               | 59,720                       | 57,660               | 56,786               |
| R <sup>2</sup>                 | .12                                                  | .13                          | .13                  | .13                  |

The dependent variable in column 1 is the credit growth differential between cross-border and domestic nonbank borrowers. The dependent variable in columns 2 to 4 is the cross-border credit flows to borrowers in all sectors, to banks, or to nonbanks. The sample period is from 1995:Q1 to 2014:Q2. The sample includes all reporting and counterparty countries listed in Table A1 in the appendix. The appendix defines all variables. Standard errors are clustered at the counterparty and reporting country levels. \* $p < .1$ ; \*\* $p < .05$ ; \*\*\* $p < .01$ .

**Table 7**  
Cross-border credit flows and monetary policy in euro area source countries

|                                     | Full sample                                    |                     | Excl. FR, GE                                   |                     |
|-------------------------------------|------------------------------------------------|---------------------|------------------------------------------------|---------------------|
|                                     | Credit growth diff., cross-border vs. domestic |                     | Credit growth diff., cross-border vs. domestic |                     |
|                                     | (1)                                            | (2)                 | (3)                                            | (4)                 |
|                                     |                                                |                     |                                                |                     |
| Lag policy rate rep                 | 0.382***<br>[0.113]                            | 0.411***<br>[0.114] | 0.303***<br>[0.099]                            | 0.433***<br>[0.116] |
| Lag policy rate rep × Euro area rep | 0.719***<br>[0.280]                            | 0.850***<br>[0.303] | 0.297<br>[0.281]                               | 0.218<br>[0.251]    |
| <i>Joint Euroarea rep</i>           | 0.719                                          | 0.850               | 0.594                                          | 0.650               |
| <i>t</i> -statistic                 | 2.572                                          | 2.805               | 2.812                                          | 2.297               |
| Controls rep                        | Yes                                            | Yes                 | Yes                                            | Yes                 |
| Controls rep × Euro area rep        | Yes                                            | Yes                 | Yes                                            | Yes                 |
| Counterparty × Year-quarter FE      | Yes                                            | Yes                 | Yes                                            | Yes                 |
| Observations                        | 65,302                                         | 55,129              | 66,401                                         | 67,018              |
| R <sup>2</sup>                      | .11                                            | .12                 | .12                                            | .11                 |

The dependent variable in columns 1 and 2 is the credit growth differential between cross-border and domestic nonbank borrowers. The dependent variable in columns 3 and 4 is the cross-border credit flows to borrowers in all sectors, to banks, or to nonbanks. *Euroarea rep* takes the value of one for Euro area reporting countries and zero otherwise. *Joint Euroarea rep* is the sum of coefficients for *Lagpolicyrate rep* and *Lagpolicyrate rep × Euroarea rep*. The sample period is from 1995:Q1 to 2014:Q2. The sample in columns 1 and 3-5 includes all reporting and counterparty countries listed in Table A1 in the appendix. The sample in columns 2 and 4 excludes Germany and France as reporting countries. The appendix defines all variables. Standard errors are clustered at the counterparty and reporting country levels. \* $p < .1$ ; \*\* $p < .05$ ; \*\*\* $p < .01$ .

**Economic message.** Funding stress reduces cross-border lending, as expected, but the monetary-policy risk-taking effect remains after controlling for funding costs.

## 5.8 Table 7: euro-area source countries

**Read:** Table 7 uses the euro area as a robustness setting. Because the ECB sets a common monetary policy for members whose domestic conditions differ, this test helps separate monetary policy from individual-country macroeconomic conditions.

**Economic message.** The positive monetary-policy effect persists for euro-area source countries and when France and Germany are excluded. This weakens the argument that the result is only a domestic-business-cycle artifact.

## 5.9 Table 8: pre- and post-GFC periods

**Read:** Table 8 separates the pre-GFC and post-GFC periods and also uses shadow policy rates for the post-GFC period. The positive effect is strongest and clearest in the pre-GFC period, while the

post-GFC evidence is more sensitive to specification and the effective lower bound.

**Table 8**  
Cross-border credit flows in the pre- and post-GFC periods

|                                | Pre-GFC (until 2007:Q2)      |                      |                      | Post-GFC (2009:Q3 onward)    |                      |                      | Post-GFC (2009:Q3 onward)    |                      |                      |
|--------------------------------|------------------------------|----------------------|----------------------|------------------------------|----------------------|----------------------|------------------------------|----------------------|----------------------|
|                                | Cross-border credit flows to |                      |                      | Cross-border credit flows to |                      |                      | Cross-border credit flows to |                      |                      |
|                                | All sectors                  | Banks                | Nonbanks             | All sectors                  | Banks                | Nonbanks             | All sectors                  | Banks                | Nonbanks             |
|                                | (1)                          | (2)                  | (3)                  | (4)                          | (5)                  | (6)                  | (7)                          | (8)                  | (9)                  |
| Lag policy rate rep            | 0.304***<br>[0.105]          | 0.272*<br>[0.129]    | 0.542***<br>[0.198]  | 0.140<br>[0.184]             | -0.099<br>[0.218]    | 0.441<br>[0.324]     |                              |                      |                      |
| Lag shadow policy rate rep     |                              |                      |                      |                              |                      |                      | 0.198<br>[0.170]             | 0.093<br>[0.292]     | 0.403*<br>[0.234]    |
| Lag credit growth rep          | 0.044<br>[0.042]             | 0.106<br>[0.070]     | 0.041<br>[0.051]     | 0.070**<br>[0.031]           | 0.034<br>[0.078]     | 0.032<br>[0.043]     | 0.000<br>[0.074]             | 0.071***<br>[0.011]  | 0.039<br>[0.039]     |
| Lag bank equity returns rep    | -0.013<br>[0.020]            | 0.001<br>[0.016]     | -0.008<br>[0.015]    | -0.006<br>[0.015]            | -0.011<br>[0.041]    | -0.021<br>[0.022]    | -0.010<br>[0.015]            | -0.020<br>[0.032]    | -0.038<br>[0.022]    |
| Lag real GDP growth rep        | 0.029<br>[0.191]             | 0.273<br>[0.417]     | -0.032<br>[0.225]    | 0.633***<br>[0.247]          | 0.042<br>[0.430]     | 1.112***<br>[0.270]  | 0.634***<br>[0.243]          | 0.066<br>[0.401]     | 1.118***<br>[0.266]  |
| Lag debt/GDP rep               | -0.008<br>[0.008]            | -0.017<br>[0.016]    | 0.001<br>[0.010]     | -0.004<br>[0.004]            | -0.024***<br>[0.008] | -0.004<br>[0.007]    | -0.000<br>[0.006]            | -0.023***<br>[0.009] | 0.002<br>[0.010]     |
| Lag inflation rep              | 0.923***<br>[0.420]          | 1.462<br>[1.013]     | 0.837<br>[0.620]     | -0.668<br>[0.509]            | -0.179<br>[0.801]    | 0.117<br>[0.608]     | -0.670<br>[0.518]            | -0.301<br>[0.797]    | 0.099<br>[0.604]     |
| Exchange rate pct. change      | -0.018<br>[0.038]            | -0.025<br>[0.066]    | -0.006<br>[0.056]    | 0.000<br>[0.028]             | 0.064<br>[0.072]     | -0.024<br>[0.044]    | 0.067<br>[0.026]             | -0.021<br>[0.070]    | -0.021<br>[0.042]    |
| Financial center rep           | -1.323*<br>[0.714]           | -4.423***<br>[1.019] | -1.759**<br>[0.836]  | -0.226<br>[0.409]            | -1.888<br>[1.210]    | 0.204<br>[0.569]     | -0.012<br>[0.758]            | -1.355<br>[1.259]    | 0.488<br>[1.073]     |
| Euro area rep                  | -0.812*<br>[0.444]           | -1.384<br>[1.179]    | -1.744***<br>[0.600] | -2.462***<br>[0.529]         | -2.794***<br>[0.807] | -2.143***<br>[0.641] | -1.328***<br>[0.536]         | -2.469***<br>[0.825] | -1.831***<br>[0.588] |
| QE indicator rep               | 2.056***<br>[0.560]          | -1.425<br>[1.484]    | 1.670**<br>[0.799]   | 0.832<br>[1.211]             | 0.884<br>[2.173]     | -0.465<br>[1.691]    | 1.151<br>[1.167]             | 1.023<br>[2.089]     | 0.306<br>[1.443]     |
| EMU rep                        | 0.365<br>[0.378]             | 0.817<br>[2.127]     | -0.268<br>[1.437]    | 1.644<br>[1.096]             | 4.962***<br>[1.172]  | -0.078<br>[1.578]    | 1.376<br>[1.019]             | 4.132***<br>[1.896]  | -0.162<br>[1.492]    |
| Counterparty × Year-quarter FE | Yes                          | Yes                  | Yes                  | Yes                          | Yes                  | Yes                  | Yes                          | Yes                  | Yes                  |
| Observations                   | 38,638                       | 37,476               | 37,910               | 21,051                       | 20,230               | 20,555               | 21,051                       | 20,230               | 20,555               |
| R <sup>2</sup>                 | .12                          | .12                  | .11                  | .12                          | .10                  | .11                  | .12                          | .10                  | .10                  |

The dependent variable consists of the cross-border credit flows to borrowers in all sectors, to banks, or to nonbanks. The sample period in columns 1–3 is from 1995:Q1 to 2007:Q2, and in columns 4–9 from 2009:Q3 to 2014:Q2. The sample includes all reporting and counterparty countries listed in Table A1 in the appendix. The appendix defines all variables. Standard errors are clustered at the counterparty and reporting country levels. \*p < .1; \*\*p < .05; \*\*\*p < .01.

**Table 9**  
Cross-border credit flows and global factors

|                              | Cross-border credit flows to |                      |                      | Cross-border credit flows to |                      |                      | Cross-border credit flows to |                      |                      |
|------------------------------|------------------------------|----------------------|----------------------|------------------------------|----------------------|----------------------|------------------------------|----------------------|----------------------|
|                              | All sectors                  |                      |                      | All sectors                  |                      |                      | All sectors                  |                      |                      |
|                              | (1)                          | (2)                  | (3)                  | (4)                          | (5)                  | (6)                  | (7)                          | (8)                  | (9)                  |
| Lag policy rate rep          | 0.210**<br>[0.086]           | 0.274*<br>[0.144]    | 0.170<br>[0.139]     | 0.188**<br>[0.085]           | 0.293**<br>[0.141]   | 0.286**<br>[0.135]   |                              |                      |                      |
| Lag policy rate cp           | -0.094**<br>[0.040]          | -0.231***<br>[0.078] | -0.059<br>[0.046]    | -0.096**<br>[0.059]          | -0.234***<br>[0.076] | -0.038<br>[0.045]    |                              |                      |                      |
| ln(VIX)                      |                              |                      |                      | -4.946***<br>[0.673]         | -4.991***<br>[1.164] | -5.783***<br>[0.729] | -4.871***<br>[0.668]         | -4.949***<br>[1.153] | -5.092***<br>[0.724] |
| Lag policy rate differential |                              |                      |                      |                              |                      |                      | 0.166**<br>[0.075]           | 0.159***<br>[0.067]  | 0.246***<br>[0.077]  |
| Lag credit growth rep        | 0.076***<br>[0.015]          | 0.094***<br>[0.026]  | 0.064***<br>[0.017]  | 0.057***<br>[0.013]          | 0.076***<br>[0.024]  | 0.050***<br>[0.013]  | 0.099***<br>[0.014]          | 0.097***<br>[0.013]  | 0.091***<br>[0.014]  |
| Lag credit growth cp         | 0.081***<br>[0.014]          | 0.122***<br>[0.024]  | 0.056***<br>[0.013]  | 0.072***<br>[0.013]          | 0.116***<br>[0.022]  | 0.041***<br>[0.013]  | 0.072***<br>[0.013]          | 0.116***<br>[0.022]  | 0.041***<br>[0.013]  |
| Lag bank equity returns rep  | -0.009<br>[0.010]            | 0.001<br>[0.018]     | -0.012<br>[0.012]    | -0.006<br>[0.008]            | -0.001<br>[0.015]    | -0.016<br>[0.010]    | -0.006<br>[0.008]            | -0.001<br>[0.015]    | -0.016<br>[0.010]    |
| Lag bank equity returns cp   | 0.014<br>[0.009]             | 0.016**<br>[0.016]   | 0.004<br>[0.009]     | 0.016**<br>[0.008]           | 0.022*<br>[0.013]    | 0.000<br>[0.008]     | 0.016**<br>[0.008]           | 0.022*<br>[0.013]    | 0.000<br>[0.008]     |
| Lag real GDP growth rep      | 0.342<br>[0.151]             | 0.270<br>[0.297]     | 0.444***<br>[0.171]  | 0.208<br>[0.130]             | 0.228<br>[0.257]     | 0.367***<br>[0.151]  | 0.214*<br>[0.130]            | 0.232<br>[0.257]     | 0.147**<br>[0.150]   |
| Lag real GDP growth cp       | 0.266**<br>[0.127]           | 0.355<br>[0.228]     | 0.266**<br>[0.141]   | 0.292**<br>[0.113]           | 0.366**<br>[0.208]   | 0.232*<br>[0.131]    | 0.269***<br>[0.113]          | 0.367*<br>[0.208]    | 0.248*<br>[0.112]    |
| Lag debt/GDP rep             | -0.017**<br>[0.007]          | -0.014<br>[0.013]    | -0.022***<br>[0.009] | -0.019***<br>[0.007]         | -0.017<br>[0.003]    | -0.025***<br>[0.009] | -0.018<br>[0.007]            | -0.016<br>[0.003]    | -0.024***<br>[0.009] |
| Lag debt/GDP cp              | -0.009<br>[0.007]            | 0.029**<br>[0.011]   | -0.040***<br>[0.011] | -0.011<br>[0.003]            | 0.026**<br>[0.011]   | -0.044***<br>[0.008] | -0.011<br>[0.003]            | 0.026**<br>[0.011]   | -0.044***<br>[0.008] |
| Lag inflation rep            | -0.012<br>[0.130]            | -0.149<br>[0.533]    | -0.178<br>[0.328]    | 0.918*<br>[0.495]            | 0.178<br>[0.297]     | 0.177<br>[0.276]     | 0.917*<br>[0.276]            | -0.303<br>[0.495]    | 0.177<br>[0.276]     |
| Lag inflation cp             | 0.276<br>[0.154]             | 0.143<br>[0.366]     | 0.297<br>[0.162]     | 0.722**<br>[0.166]           | 0.369<br>[0.347]     | 0.162<br>[0.162]     | 0.794**<br>[0.162]           | 0.382<br>[0.348]     | 0.184<br>[0.161]     |
| Exchange rate pct. change    | -0.016<br>[0.015]            | -0.015<br>[0.025]    | -0.012<br>[0.017]    | -0.011<br>[0.025]            | -0.003<br>[0.017]    | -0.003<br>[0.017]    | -0.011<br>[0.025]            | -0.003<br>[0.017]    | -0.003<br>[0.025]    |
| Euro area rep                | -0.118<br>[0.760]            | -2.391***<br>[1.324] | -0.144<br>[1.051]    | 0.617<br>[1.973]             | -2.255***<br>[1.051] | -0.154<br>[0.757]    | 0.612<br>[1.973]             | -2.233***<br>[1.316] | 0.612<br>[1.973]     |
| QE indicator rep             | 1.977***<br>[0.550]          | 2.830***<br>[0.673]  | 2.649***<br>[0.551]  | 1.862***<br>[0.551]          | 2.767***<br>[0.920]  | 2.661***<br>[0.670]  | 1.751***<br>[0.540]          | 2.768***<br>[0.899]  | 2.514***<br>[0.652]  |
| Counterparty × Reporting FE  | Yes                          | Yes                  | Yes                  | Yes                          | Yes                  | Yes                  | Yes                          | Yes                  | Yes                  |
| Year-quarter FE              | Yes                          | Yes                  | Yes                  | Yes                          | Yes                  | Yes                  | Yes                          | Yes                  | Yes                  |
| Observations                 | 42,664                       | 41,973               | 41,688               | 42,664                       | 41,973               | 41,688               | 42,664                       | 41,973               | 41,688               |
| R <sup>2</sup>               | .04                          | .04                  | .03                  | .03                          | .03                  | .03                  | .03                          | .03                  | .03                  |

The dependent variable consists of the cross-border credit flows to borrowers in all sectors, to banks, or to nonbanks. The sample period is from 1995:Q1 to 2014:Q2. The sample includes all reporting and counterparty countries listed in Table A1 in the appendix. The appendix defines all variables. Standard errors are clustered at the counterparty and reporting country levels. \*p < .1; \*\*p < .05; \*\*\*p < .01.

**Economic message.** The mechanism is easiest to detect during conventional monetary-policy periods. After the crisis, zero lower bound constraints, QE, and changing banking regulation complicate the interpretation.

## 5.10 Table 9: global factors

**Read:** Table 9 controls for reporting-counterparty fixed effects and either time fixed effects or explicit global factors such as ln(VIX). Source-country policy remains positive in several specifications, destination-country policy tends to be negative, and the VIX has a large negative effect on cross-border flows.

**Economic message.** Global risk aversion matters, but it does not eliminate the importance of source-country monetary policy. Local monetary policy and global financial conditions operate simultaneously.

## 5.11 Appendix Table A1: reporting and counterparty countries

**Read:** Appendix Table A1 lists the reporting/source and counterparty/destination countries in the sample, along with observations and emerging-market status.

**Economic message.** The country coverage is broad. This breadth is what permits the paper to study monetary-policy transmission across many source-country banking systems rather than focusing only on the United States.

# 6 Short summary

- The paper studies whether monetary policy affects the international composition of bank credit.
- The key data are BIS locational banking statistics paired with newly constructed domestic bank-credit series.



**Table A1**  
**List of reporting and counterparty countries**

*A. Reporting/source countries*

| Reporting country | Obs.  | EME status | Reporting country | Obs.   | EME status |
|-------------------|-------|------------|-------------------|--------|------------|
| Australia         | 1,324 | 0          | Japan             | 4,015  | 0          |
| Austria           | 3,616 | 0          | Korea             | 1,920  | 0          |
| Belgium           | 3,818 | 0          | Luxembourg        | 2,322  | 0          |
| Brazil            | 741   | 1          | Malaysia          | 722    | 1          |
| Canada            | 2,205 | 0          | Mexico            | 149    | 1          |
| Denmark           | 2,022 | 0          | Netherlands, The  | 3,394  | 0          |
| Finland           | 1,390 | 0          | Portugal          | 1,379  | 0          |
| France            | 4,952 | 0          | South Africa      | 224    | 1          |
| Germany           | 5,006 | 0          | Spain             | 3,056  | 0          |
| Greece            | 749   | 0          | Sweden            | 2,105  | 0          |
| Hong Kong         | 2,016 | 0          | Switzerland       | 4,964  | 0          |
| India             | 1,620 | 1          | Turkey            | 670    | 1          |
| Indonesia         | 189   | 1          | United Kingdom    | 4,964  | 0          |
| Ireland           | 2,129 | 0          | United States     | 3,683  | 0          |
| Italy             | 3,059 | 0          | Total             | 68,403 | 7 of 29    |

*B. Counterparty/destination countries*

| Counterparty country | Obs.  | EME status | Counterparty country | Obs.   | EME status |
|----------------------|-------|------------|----------------------|--------|------------|
| Algeria              | 431   | 1          | Lithuania            | 254    | 1          |
| Argentina            | 955   | 1          | Luxembourg           | 1,391  | 0          |
| Australia            | 1,224 | 0          | Malaysia             | 875    | 1          |
| Austria              | 1,299 | 0          | Mauritius            | 354    | 1          |
| Belgium              | 1,393 | 0          | Mexico               | 1,149  | 1          |
| Bolivia              | 82    | 1          | Morocco              | 841    | 1          |
| Brazil               | 1,198 | 1          | Netherlands, The     | 1,511  | 0          |
| Bulgaria             | 619   | 1          | New Zealand          | 845    | 0          |
| Canada               | 1,303 | 0          | Norway               | 1,297  | 0          |
| Chile                | 1,100 | 1          | Oman                 | 483    | 1          |
| China                | 1,278 | 1          | Pakistan             | 669    | 1          |
| Colombia             | 660   | 1          | Panama               | 1,026  | 1          |
| Cote d'Ivoire        | 219   | 1          | Paraguay             | 316    | 1          |
| Croatia              | 424   | 1          | Peru                 | 870    | 1          |
| Cyprus               | 713   | 0          | Philippines          | 945    | 1          |
| Czech Republic       | 895   | 0          | Poland               | 1,060  | 1          |
| Denmark              | 1,305 | 0          | Portugal             | 1,212  | 0          |
| Estonia              | 78    | 0          | Qatar                | 537    | 1          |
| Finland              | 1,194 | 0          | Romania              | 596    | 1          |
| France               | 1,519 | 0          | Russia               | 1,224  | 1          |
| Germany              | 1,484 | 0          | Saudi Arabia         | 938    | 1          |
| Ghana                | 307   | 1          | Senegal              | 158    | 1          |
| Greece               | 1,070 | 0          | Singapore            | 1,384  | 1          |
| Guatemala            | 324   | 1          | Slovak Republic      | 523    | 0          |
| Hong Kong            | 1,273 | 0          | Slovenia             | 546    | 0          |
| Hungary              | 882   | 1          | South Africa         | 1,115  | 1          |
| Iceland              | 793   | 0          | Spain                | 1,318  | 0          |
| India                | 1,003 | 1          | Sri Lanka            | 512    | 1          |
| Indonesia            | 1,225 | 1          | Sweden               | 1,301  | 0          |
| Ireland              | 1,403 | 0          | Switzerland          | 1,479  | 0          |
| Israel               | 958   | 1          | Taiwan               | 883    | 1          |
| Italy                | 1,408 | 0          | Thailand             | 882    | 1          |
| Jamaica              | 219   | 1          | Tunisia              | 591    | 1          |
| Japan                | 1,435 | 0          | Turkey               | 1,228  | 1          |
| Jordan               | 397   | 1          | Ukraine              | 287    | 1          |
| Korea                | 1,063 | 0          | United Kingdom       | 1,538  | 0          |
| Kuwait               | 534   | 1          | United States        | 1,533  | 0          |
| Libya                | 157   | 1          | Venezuela            | 908    | 1          |
|                      |       |            | Total                | 68,403 | 16 of 76   |

- The empirical strategy compares flows from different source countries to the same destination country in the same quarter.
- A tighter source-country monetary stance is associated with faster cross-border credit growth relative to domestic credit growth.
- The effect is stronger for lower-capitalized or more stressed banking sectors.
- The effect is directed mainly toward safer foreign destinations.
- Robustness tests address funding costs, euro-area identification, pre/post-GFC differences, shadow rates, VIX/global factors, and destination-country policy rates.

## 7 Conclusion

### 7.1 Core conclusion

Correa, Paligorova, Saprizza, and Zlate show that source-country monetary policy shapes global banks' international credit portfolios. A relatively tighter source-country policy rate is associated with more cross-border credit relative to domestic credit, especially toward safer foreign destinations.

### 7.2 Economic intuition

Tighter monetary policy increases the relative riskiness of domestic borrowers by weakening collateral values and balance sheets. Banks respond by shifting their portfolios toward foreign borrowers that are relatively safer. This is the cross-border version of the risk-taking channel.

### 7.3 Takeaway

The paper's main message is that monetary policy has international portfolio effects. Policymakers in source countries should expect domestic monetary tightening to affect banks' foreign exposures, while policymakers in recipient countries should monitor source-country monetary conditions as a determinant of foreign bank credit supply.